FAST School of Computing, NUCES Islamabad

Granite

find the failure · prove the cause · verify the fix

A Self-Healing Multi-Agent System for Semantic Error Diagnosis, Verified Repair and Recovery

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PROJECT GOALS

- Detect silent failures in agent workflows
- Locate where the error began
- Prove the cause by controlled replay
- Keep a person in charge, with evidence

UN SDG 9 · Industry, Innovation and Infrastructure

01 · THE PROBLEM

It looked fine. It was wrong.

- Agents pass a wrong value along silently
- The run completes; no error is raised
- Blame lands where it showed, not where it began
- One lucky re-run can pass a wrong fix

Two views, same run

What output monitoring sees

- Run status: completed
- Errors raised: none
- Reply: refund of $662.00

Looks healthy

Checked against the catalogue

What Granite's check sees

Told to the customer $662.00 · Catalogue total $662.23

Difference -0.23 · CHK-OUT-1 failed

Refund Desk run, recorded mode. **Silent failure found**

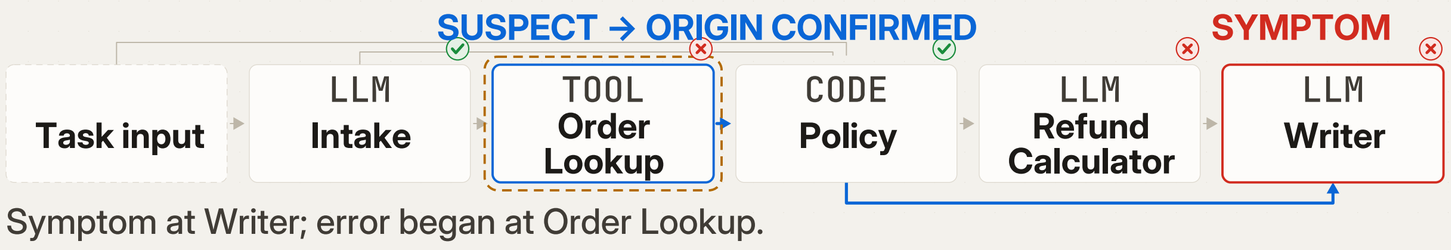
02 · THE METHOD

Seven steps, four stages, one record

V0 Detected · V1 Traced · V2 Verified · V3 Reviewed

Run → Check → Detect → Trace back → Verify → Review → Record

Where did it begin?



Symptom at Writer; error began at Order Lookup.

Three ideas

Executable checks
Rules test each step against a trusted source.

Deterministic attribution
Suspects ranked from the recorded data flow.

Controlled replay
Fix replays vs control replays, under a fixed rule.

03 · THE PROOF

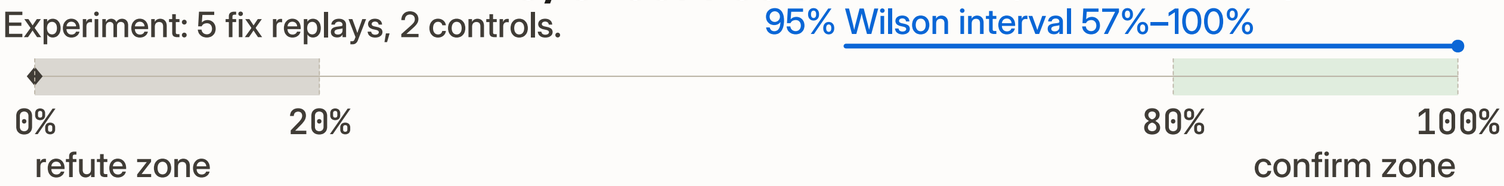
One re-run proves nothing. Controls do.

Order Lookup → CONFIRMED

With the fix 5/5 correct · Control, no fix 0/2 correct

Experiment: 5 fix replays, 2 controls.

95% Wilson interval 57%–100%



Refund Calculator → REFUTED

With the fix 0/5 correct · Control 0/2 correct

Rule VR-1

- Confirmed: ≥ 80% fix correct, ≤ 20% controls correct
- Refuted: ≤ 20% fix correct
- Otherwise: inconclusive

Rule fixed before any result

SHA-256 hashed evidence
A person decides

04 · BUILT AND MEASURED

A working product, measured honestly

Pre-registered feasibility run (7 Oct 2026)

10/10 faulty runs detected	0 false alarms on clean runs*	10/10 origin ranked first
9/10 origin confirmed by replay	10/10 innocent step refuted	20/20 runs completed

*A genuine model error, not injected. 10 tasks, 1 fault type, 1 small model: a first test, not a benchmark.

Engineering

- Automated tests
- CI/CD on GitHub Actions
- Live on Render
- 21 diagrams from the code

05 · PLAN

Plan to the FYP-1 final evaluation

Planned

- Fix values derived from failed checks
- One-click verified investigation
- Three fault types and a batch Fault Lab
- Evaluation against baselines
- Trace import
- Regression-test export
- Retail Agent on tau-bench tasks